Pilot predictions

Tom · 30 Oct 2026 · committed before the pilot run

Setup: 64×64 greyscale images, centred 16×16 gap (the centre is 8 pixels from the nearest visible pixel). Reach: networks 1–2 whole image; 3–4 16 pixels out; 5–6 12 out; 7–8 0.

Predicted mechanism that dominates: continuation (gap filled by continuing what's next to it; Hume's contiguity), **resemblance** (a pattern from elsewhere carried into the gap by shared filters; Hume's resemblance), **neither** (no sign of the surrounding image) or **can't tell** (both give the same fill). Judged by eye using the key above each table.

"About" and "beats": "about X" means a gap MSE within 20% of X's; "beats X" / "better than X" means more than 20% lower; "worse than X" means more than 20% higher.

Observed rows: rows marked (observed) were already seen in a small test on 29 Oct (8 channels, 400 steps, 1 seed), so they aren't blind. The real tests are networks 3, 4, 7 and 8, every blurred-noise row, and the borrowed rows.

Borrowed rows: network 2 is trained as normal, then each gap position is given the trained last-layer filters (last 3×3, 1×1 and output layer) of a donor visible position: the **nearest** one, or the one whose arriving features are **most similar**. This tests whether useful information from the rest of the image reaches the gap through the coarse summary, independently of weight sharing. The donor choice forces the mechanism, so the bet is whether it works.

Gradient

Needs only nearness; harmonic fill is exact here, so "near perfect" is the best any network can do.

> **Key.** Continuation and resemblance both give a smooth ramp here, so the only possible calls are can't tell (a ramp appears) or neither (no ramp).

Net	Outcome	Predicted mechanism that dominates	Reason	Confidence
1	Near perfect **(observed)**	Can't tell	Coarse cells shared with the surroundings give a smooth continuation	High
2	Worse than grey **(observed)**	Neither	Last layers at gap positions never trained, so arriving information is scrambled	High
3	Near perfect	Can't tell	Upsampling alone gives a smooth ramp, which is exactly what a linear gradient is	Medium
4	About grey or worse	Neither	Unshared: the gap's last layers are untrained	High
5	Worse than grey **(observed, small test)**	Can't tell near the edges; neither in the centre	No coarse summary; edge information creeps in only locally; short training	Low
6	Worse than grey **(observed)**	Neither	Unshared: the gap's last layers are untrained	High
7	Worse than grey: gap MSE about 2× grey's	Neither	No links between pixels; the gap is filled from its own noise	High
8	Worse than grey	Neither	Nothing spatial, nothing trained in the gap	High
2 + borrowed (nearest donor)	Between harmonic and grey; better than plain network 2	Continuation by design	The coarse summary arrives; the nearest edge's trained filters render it as a smooth continuation	Low
2 + borrowed (most-similar donor)	About the same as the nearest donor	Resemblance by design	The gradient has no repeated pattern, so the most similar positions are effectively nearby ones	Low

Blurred noise (ℓ = 4)

Needs nearness at scale 4: the edges help near the border, little in the centre. The exact best-possible fill is known, so no network can beat it on average.

> **Key.** Continuation: smooth blobs carried on from the edges, fading towards the centre. Resemblance: blob texture across the whole gap, copied from elsewhere, possibly not matching the edges. Neither: speckle or static. This image separates the two only weakly.

Net	Outcome	Predicted mechanism that dominates	Reason	Confidence
1	Beats harmonic, well short of the best possible	Continuation (no repeated blobs to copy)	Coarse cells capture blob structure at several scales; the centre stays uncertain	Medium
2	About grey or worse	Neither	Unshared: arriving information scrambled	High
3	About harmonic	Continuation	Smooth coarse continuation, no fine detail	Medium
4	About grey or worse	Neither	Unshared	High
5	Between harmonic and grey	Continuation near the edges; neither in the centre	Edge information reaches about 12 pixels; centre weakly informed; no coarse summary	Medium
6	Worse than grey	Neither	Unshared	High
7	Worse than grey: gap MSE about 2× grey's	Neither	No links between pixels	High
8	Worse than grey	Neither	Nothing spatial	High
2 + borrowed (nearest donor)	Better than plain network 2; about harmonic	Continuation by design	The nearest edge's filters render the arriving coarse summary as a smooth carry-on from the edge, which is what blurred noise needs	Low
2 + borrowed (most-similar donor)	Better than plain network 2; worse than the nearest donor	Resemblance by design	Blobs don't repeat, so a similar-looking donor elsewhere brings texture that doesn't match this gap's edges	Low

Stripes

Needs a pattern carried from elsewhere, in the right place. Harmonic fill gives a grey smear (gap MSE 0.125, the same as grey). This is where continuation and resemblance split.

> **Key.** Resemblance: stripes visible across the middle of the gap at the right angle, even if out of phase. Continuation: stripes fade within a few pixels of the edge, and the centre is a smear near the edge average. Neither: noise or unrelated structure.

Net	Outcome	Predicted mechanism that dominates	Reason	Confidence
1	Worse than grey **(observed, small test)**: possibly stripes out of phase	Resemblance	Shared filters can draw stripe texture, but the phase in the centre is uncertain; small test only	Low
2	About grey **(observed)**	Neither	Unshared: scrambled	High
3	About grey	Continuation	1×1 filters and coarse cells only, so it can't render period-8 stripes: a smooth blur	Medium
4	About grey or worse	Neither	Unshared	High
5	Beats harmonic **(observed, small test)**	Resemblance	Shared 3×3 filters draw stripes in from the edges (reach 12 > 8)	Medium
6	About grey **(observed)**	Neither	Unshared	High
7	Worse than grey: gap MSE about 2× grey's	Neither	No links between pixels	High
8	Worse than grey	Neither	Nothing spatial	High
2 + borrowed (nearest donor)	About grey to slightly better; better than plain network 2	Continuation by design	The nearest edge's filters render the arriving coarse summary as a smooth continuation; no pattern is reused	Low
2 + borrowed (most-similar donor)	Beats the nearest donor; some stripes, phase uncertain; still short of network 1	Resemblance by design	The donor is chosen by matching features, so its filters come from a similar part of the stripe pattern	Low